



Coherent Heuristic Evaluation (CoHE): Toward Increasing the Effectiveness of Heuristic Evaluation for Novice Evaluators

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Abstract. Heuristic evaluation (HE) is an inspection-based usability evaluation method in which a number of evaluators, typically 3–5, assess the usability of a system based on a set of usability guidelines. HE was first introduced by Nielsen and Molich and then revised by Nielsen. Since its introduction, HE has gained wide popularity among human-computer interaction (HCI) and user experience (UX) practitioners and is now one of the most common usability evaluation methods. A few years after the introduction of HE, some researchers realized that novice evaluators perform poorly on complex systems. This issue, known as the expertise effect, is one of the major challenges of HE; i.e., the more experienced the evaluator is, the better the results of the evaluation. Consequently, some researchers argue that the results of HE are the product of the evaluator's experience rather than the method itself. To address this issue, we interviewed 15 usability experts, all of whom had at least four years of experience and who came from both academia and industry. We analyzed their responses and developed a step-by-step protocol.

Keywords: Usability · Inspection-based methods · Heuristic Evaluation · Novice evaluators

1 Introduction

Technology is found everywhere around us, from smart phones to smart watches to tablets, and is involved in every aspect of our lives, from communication to education to health. It has basically changed the way we live, and any person in the present day must interact with it. This was not the case until two decades ago. Before then, technology was limited to certain groups of tech-savvy people. However, the focus on developing systems that are merely functional is no longer sufficient; systems must be usable enough to be used by ordinary people who are not tech savvy. Usability is one of the main objectives of the field of human-computer interaction (HCI) [1]. To increase the usability of systems, a number of methods have been proposed, one of which is heuristic evaluation (HE). HE is a method in which a number of evaluators, typically 3–5, assess a system based on a set of usability guidelines. HE was first developed and proposed by Nielsen and Molich [2, 3] and then revised by Nielsen [4]. Currently, HE is very popular and is one of the most common usability evaluation

methods [5, 6]. Despite its popularity, HE suffers from multiple problems to the extent that some researchers have started to question its reliability [7]. One of the major issues of HE is the expertise effect [8], which means that the more experienced the evaluator is, the better the results of the evaluation [9]. This issue has led some researchers to say that the results of HE are the product of the evaluator's experience rather than of HE as a method [7]. A potential reason for this problem is that HE is not a structured method [7], which means there is no step-by-step guide on how to perform it. This lack of direction is especially problematic for novice evaluators who have no or very limited experience. Moreover, HCI is a relatively new field, so there is a scarcity of HCI experts. Even in research projects, researchers depend on novice evaluators to perform HE [10, 11]. Even if experts are available, hiring them is usually expensive, which explains why startups and small companies depend on novice evaluators [12]. HCI is a multidisciplinary field, and HCI practitioners come from different backgrounds, such as computer science, psychology, art, and business; some have no formal HCI education [13]. Ideally, potential users should be involved in the development life cycle. However, HCI practitioners are sometimes expected to represent the user [14], which increases the importance of their work. Additionally, HE should ideally be performed by 3–5 evaluators to increase the likelihood of detecting most usability problems. However, some companies hire a single HCI practitioner, as shown in a survey conducted in Malaysia [15]. Poor usability has large economic costs [16] and can lead to serious health and safety risks [17–19]. Therefore, there is a need to develop a step-by-step protocol to guide novice evaluators through HE to improve the quality of their evaluations, thus improving the usability of the systems that they are evaluating. In this study, our aim is to create such a protocol. We interviewed 15 usability experts from both academia and industry about their experience with HE, the difficulties they faced with it, how they overcame these difficulties, and their suggestions on how to improve HE for novices. We coded and synthesized their responses and created a coherent, step-by-step protocol for HE.

2 Related Work

A review of the HE literature indicates four directions. The first direction focuses on producing general usability guidelines. This direction started even before the proposal of HE in 1990 by Nielsen and Molich [2, 3]. For example, in 1987, Shneiderman [20] produced eight golden rules based on his experience in the field; these rules are general guidelines meant to help designers in designing their system and assessing its usability. After Nielsen's proposal of HE and the ten usability heuristics, many researchers and usability experts tried to follow suit by providing their own usability heuristics and guidelines. These guidelines differ; some are more abstract than others, and there are some overlaps between them. However, most of them are driven by experts' experience in the field. To name a few, Tognazzini [21] created a longer list that contains 19 principles, and Gerhardt-Powals [22] produced a list of 10 principles through her work in cognitive science. A less well-known list is Mandel's [23] three golden rules. Due to the abundance of usability guidelines, some researchers began to compare them or even merge them. For example, in [24], the author merged Nielsen's ten heuristics with

Tognazzini's principles to develop a new set of 15 heuristics, and in [25], the author compared Nielsen's ten heuristics to Gerhardt-Powals's 10 principles.

In contrast to the first direction, the second direction focuses on producing domain-specific heuristics. Many researchers believed that the usability guidelines were very general and difficult to implement in certain domains. Therefore, they started to produce heuristics tailored to specific audiences, platforms, disciplines and contexts. Examples are [26], which is a list of 12 usability heuristics for touch-screen mobile devices; [27], which is a list of 18 usability heuristics for automated teller machines (ATMs); [28], which presents 11 sets of heuristics to evaluate usability and the overall user experience (UX) of social networks; and [29], which generated a set of heuristics for mobile learning applications intended to be used by a Malaysian audience. The aforementioned studies are just examples of the work that has been performed in this area, and many other papers have been published with domain-specific heuristics in areas such as healthcare systems, e-government, and video games.

The aforementioned directions focus on developing a set of usability heuristics. However, usability heuristics are only one component of the HE process; HE as a method needs to be enhanced as well. This need has been realized by a number of HCI researchers. Therefore, researchers pursuing the third direction started to re-evaluate the effectiveness of the method and attempt to improve its various elements to help evaluators utilize it better. They worked on improving the understanding of the guidelines, finding better ways to implement them, and enhancing the reporting of usability issues.

Usability heuristics are abstract and relatively vague and can be open to different interpretations. The authors of [30] realized this issue and analyzed two sets of heuristics: Nielsen's heuristics and Shneiderman's principles. They developed a set of principles on how to improve usability guidelines, such as explaining the relevance of each heuristic, changing the language form of some heuristics to be active rather than passive, and simplifying some heuristics that contain more than one concept. When performing HE, the evaluator should identify usability problems that potential users of the system will encounter. To avoid finding false positive usability issues, the authors of [31] suggested including personas in the process of HE. However, when they compared the results of HE with personas with traditional HE, they found no significant difference in the results. Another way to address this issue is to conduct HE based on specific scenarios. Therefore, a method called heuristics walkthrough (HW) was developed and applied in two studies [32, 33]. The two studies compared HW to HE with contradictory results. In the first study, HW outperformed HE, while in the second study, there was no significant difference between HE and HW. The authors of [31–33] aimed to keep potential users in the mind of the evaluator during HE, but in [34], the authors suggested adding real users during the evaluation; they called this method participatory heuristic evaluation (PHE). They did not mean the addition of potential end users but rather a person with experience related to the domain being evaluated. Another technique is the hybrid usability methodology (HUM) proposed in [35], which suggests integrating HE with usability testing (UT). Combining both HE and UT increases the number of usability issues found. Performing HE in one session can be an overwhelming endeavor. The authors of [36] realized this issue and modified traditional HE to a different version called the structured heuristic evaluation method (sHEM).

The idea of sHEM is to break down the heuristic list into smaller categories and then perform the evaluation for each category in a separate session. When sHEM was tested against HE, the results showed that sHEM was superior [37]. Reporting usability issues is an important component of HE. In fact, it is arguably the most important component because it conveys the results of the evaluation. For that reason, the authors of [38] attempted to improve the formatting of the reports. Rather than using the simple format, which includes describing the usability issue and which heuristic it violated, they suggested a new format that is divided into four parts: first, describing the problem, likely/actual difficulties, the specific context and assumed causes; second, stating the discovery resources and methods; third, stating why the problem is related to a certain heuristic; and finally, stating why the problem should be eliminated. The formatting they suggested improved the quality of the usability issues detected.

HCI is a multidisciplinary and relatively new field, and HCI experts are not easily available and are expensive to hire. Therefore, in research and in industry, novice usability evaluators are used to perform HE. However, novice evaluators can produce poor results. Just a few years after the introduction of HE, the authors of [39] realized this was a problem. In their study, novice evaluators found approximately 23% of the usability issues. In an effort to understand what novice evaluators think about HE, the authors of [40] asked 31 students to express their opinion about HE after evaluating two websites. While most of the students thought it was a useful method, they thought it was difficult to implement. The results of this study, along with those of [41], provided valuable insights into the difficulties that novice evaluators face during HE. For example, novice evaluators faced some difficulties linking the usability problems to the correct heuristic. Additionally, writing the final report was considered problematic. Another effort to understand what makes HE difficult for novices is presented in [42]. In this paper, the authors noted that some novice evaluators had difficulty distinguishing between the heuristics. They suggested renaming some of the heuristics to eliminate confusion.

In contrast, expert evaluators can typically identify 74%–87% of usability issues [9], therefore there is a significant difference between the performance of experts and novices. The researchers in [43] attempted to bridge that gap by interviewing four usability experts. They showed the experts Nielsen's heuristics and asked them to talk about the tactics they used for each heuristic. Consequently, they developed a list of 38 tactics, or 3–6 tactics per heuristic. Usually, HE is performed by multiple evaluators, typically 3–5. However, each evaluator performs the evaluation separately, and they then aggregate the results. To improve the quality of HE, a method called collaborative heuristic evaluation (CHE) was developed [44] that enabled the evaluators to perform the evaluation as a group. This method was tailored for novice evaluators by [45] based on the idea of adding an expert to a group of novice evaluators to see whether the inclusion of an expert in the group increased the quality of the evaluations. When the results of the group of novices plus the expert were compared to those of a group of novices and a group of experts, the findings indicated that the results of the group of novices plus the expert were closer to those of the group of experts than to those of the group of novices. In another attempt to increase the quality of novices' evaluations, the authors of [46] suggested adding real potential users to the HE process and proposed two methods: a user exploratory session of heuristic evaluation (UES-HE) and a user

review session of heuristic evaluation (URS-HE). In the former method, users were included before the evaluation and allowed to explore the system, after which the evaluator evaluated the system. The latter method involved the opposite approach: the evaluator evaluated the system, and then the users and the evaluator reviewed the system. When UES-HE and URS-HE were compared to HE, both produced better results than HE.

The goal of this paper is to provide a step-by-step protocol on how to perform HE to facilitate the use of HE for novices, reduce the confusion they may experience, and increase the effectiveness of their evaluations. The literature on HE indicates that some researchers are concerned with developing new heuristics or combining existing heuristics, as shown in the first and second directions. However, HE is a process, and usability heuristics are only one part of it. Improving HE is also a concern, as seen in the third and fourth directions; however, most of the papers approached a particular aspect of HE rather than the entire process. Although these papers provided excellent insight, they did not address the overall issue.

3 Methodology

To develop a step-by-step protocol, we interviewed usability experts. Although this work is aimed at novices, we chose to interview usability experts because they once were novices and experienced the difficulties that novices face. Therefore, they have a good understanding of how to overcome these difficulties. To choose the participants, we first had to define an expert because what makes an individual an expert is not well defined in the literature. This lack of a definitive answer occurs not only in HCI but across all disciplines to different degrees. Some fields have higher interrater reliability (for example, weather forecasting has 95% interrater reliability), whereas other fields have lower interrater reliability (for example, clinical psychology has 40% interrater reliability) [47]. One reason for this discrepancy is that validating the outcomes is more difficult in some fields, especially those that deal with human behavior [47].

To the best of our knowledge, little work has been done to define experts in the field of HCI. The only attempt to do so is [48], in which the authors aimed to classify usability practitioners into five categories: novice, beginner, intermediate, senior and expert. They based their work on the famous idea of deliberate practice [49] and suggested that to be a usability expert, an individual should have a master's degree or PhD in the field and at least 10,000 h of usability practice, which equals 10 years of experience. These are somewhat strict criteria and are based on the idea of deliberate practice, which has been criticized. For example, the authors of [50] stated that deliberate practice is important, but alone, it does not explain how an individual becomes an expert. They noted that other important factors are involved in expertise, such as IQ, starting age, and personality.

Since it is difficult to find many usability experts with 10 years of experience and deliberate practice has been criticized, we attempted to follow previous studies in their methods of interviewing experts, particularly the approach used by the authors of [43]. Thus, we consider the minimum experience for an expert to be 4 years of experience in the field.

When choosing the number of experts to interview, interviews should typically be conducted until data saturation is achieved, meaning no new insights are added. In [51], the authors suggested that to reach saturation, 12 interviews are needed. Therefore, we planned to conduct 10–15 interviews and to add more if we did not feel that we had reached saturation. The number of interviews also depended on the availability of usability experts.

We interviewed 15 usability experts from both academia (7 participants) and industry (8 participants). Ten of them were male, and 5 were female. All of them had at least 4 years of experience, and the maximum amount of experience among the experts whom we interviewed was 15 years. The participants from academia had experience teaching HE and greater access to novices; they also should have previously performed HE at least three times. The participants from industry had more hands-on experience with the method, which gave them more knowledge of the actual techniques used. Interviewing people from both academia and industry gave us better insight into both the knowledge and the techniques behind HE because we believe that knowing the techniques without understanding the knowledge behind them would be less effective and vice versa.

The semistructured interviews were held between May 2019 and July 2019. The major items involved asking participants to describe their overall experience with HE, the difficulties they faced and how they overcame them, the detailed process of conducting HE, and how they documented usability problems, mapped usability problems to the heuristics, and estimated the severity of usability problems. They were also asked to suggest how to improve the understandability and the applicability of HE for novices. In addition, the interviewees were given the chance to talk about anything related to the topic to provide more insight. The interviewer asked questions based on the interviewees' responses.

The interviews were conducted either face-to-face (6 interviews) or via conference calls (9 interviews). After the interviewees gave their consent, each interview was audio recorded. During each interview, the interviewer took notes on the major points made by the interviewees. After 13 interviews, we felt that we had reached saturation. However, we continued our interviews to confirm that belief. After all 15 interviews were completed, the interviews were transcribed, and the audio recordings were deleted.

After the interviews, we analyzed the data to derive coherent insights. We started by examining the similarities in the participants' responses and organizing them into categories. Then, we created higher levels of categories. In this process, we focused on two things. First, we focused on the problems that the usability experts faced when they started performing HE or the mistakes they had seen novices make, how to overcome these difficulties, and how to improve HE as a method. Second, we focused on the overall organization of the HE process. Based on the analysis, we developed a step-by-step protocol.

4 Results

4.1 Problems Evaluators Faced, How to Overcome Them and Suggestions to Improve HE

1. Before performing HE, the evaluator should have some knowledge about human abilities and the way people interact with technology. Considering usability heuristics to be standalone guidelines without understanding why we apply them could lead to incorrect judgments. The reason for performing HE is to help users accomplish their goals in the easiest way possible. Therefore, two main aspects should be understood before performing HE. First, how do humans go about accomplishing their goals? Second, what are the usability components, and how will they help? Consequently, the evaluator should understand these two aspects before undertaking HE. Norman created a famous model called the seven stages of action in his book *The Design of Everyday Things* [52], which explained how humans perform tasks. Usability is a general term that has many definitions, but Nielsen provided a good definition in which he explained its components. After reading and understanding Norman's seven stages of action and Nielsen's definition of usability, the evaluators would have a better understanding of why they are performing HE.
2. Usability heuristics should help users accomplish their goals and should make the means of accomplishing the goal usable and easy. Therefore, the set of heuristics should be mapped to the seven stages of action as well as the usability components. The evaluator should understand how each of these usability heuristics helps in the process of accomplishing the goal and in which steps of the process it plays a role. The same applies to usability: in what components of usability will each heuristic help?
3. The traditional way of representing usability heuristics does not provide an effective understanding of heuristics. Simply providing a label with 2–3 lines of description and an example is not a good way to explain heuristics. There should be more information about each heuristic, and multiple aspects should be addressed:
 - Give examples that are not limited to the interface. To provide a deeper understanding of each heuristic, provide examples of how these heuristics could be used in everyday life in situations outside the interface.
 - For each heuristic, give multiple examples that are related to the interfaces; between 3 and 5 examples would highlight different issues that these heuristics address.
 - When providing examples, present both good and bad examples. Good examples help the evaluators understand what should be done, and bad examples help the evaluators understand what to avoid.
 - Explain the different components of the heuristics. Some of the heuristics deal with complex ideas that have multiple components, and these components should be clearly written. For example, Nielsen's first heuristic, "Visibility of System Status," states that the user should always be informed about the status

of the system. There are three components of this heuristic: first, the existence of the system status; second, the visibility of the system status to the user; and third, making the status visible to the user in a timely manner.

- Explain the different aspects that each heuristic addresses. For example, Nielsen's second heuristic, "Match Between System and the Real World," says that the system should speak the user's language. Clearly, this heuristic deals with multiple aspects, such as the language used, metaphors, and icons.
 - Explain the significance of each heuristic. For example, explain in detail why each heuristic is important and the consequences of ignoring it. Supporting these explanations with empirical results and psychological principles will help in emphasizing their importance.
 - Explain the applicability of each heuristic. Some evaluators think that if a system violates one of the heuristics, then it automatically indicates a usability issue that needs to be fixed. This is not the case because there are sometimes good reasons for violating the heuristics. This idea should be clear to the evaluators. Therefore, providing them with examples of situations in which violating the usability heuristics is acceptable will help clarify this idea.
 - Explain that some of the heuristics deal with more than one related concept. For example, Nielsen's tenth heuristic, "Help and Documentation," deals with two concepts: providing help to users when needed and the existence of documentation for the system. Although these two are closely related, they are not the same. Putting them into one heuristic tends to make the evaluator focus on only one of them. Separating them into two different heuristics forces the evaluator to examine each concept more carefully.
 - Some of the heuristics are labeled in a way that does not convey their meaning. Renaming these heuristics not only enhances their understandability but also helps the evaluator remember them later.
4. Deciding how to inspect the system is another issue, and there is more than one way to do so. The system could be inspected page by page, by certain tasks, by certain aspects, etc. Evaluators will sometimes be asked to inspect only one page or a very small part of the system, and in this case, their job is easier. In contrast, inspecting the entire system could be an overwhelming job for novice evaluators. To choose how to perform the inspection, evaluators should know the main services or purposes of the system. On that basis, they can develop a set of goals rather than tasks. Then, the system can be inspected according to these goals. Moreover, specific parts of the system, such as the homepage, the contact page, and the about us page, should be inspected.
 5. The timing of the inspection session is yet another problem. Some evaluators inspect the entire system in one session, which may take up to two hours or more in large systems. This might affect the evaluators' mental stamina and cause them to lose their concentration. To overcome this issue, the inspection session should be broken down into smaller sessions with a suggested duration of between 30 and 45 min. If the system is simple and can be evaluated within 30–45 min, then that duration is good. However, if the evaluation will take longer than that amount of time, breaking it down into smaller sessions is preferable.

6. Usually, evaluators are fully alert when performing the inspection, but being very focused and alert might cause some problems. Evaluators might miss certain usability issues because their performance while alert is likely to be superior. Moreover, they could become overly critical because of their strong focus. In both cases, they might report incorrect results. To address this issue, the evaluation should be performed twice so that the evaluators approach each session in a different mood: first, tired or sleepy, to resemble the situation in which real users use the system, and second, fully focused and more critical. The two sessions will help the evaluator detect different usability issues.
7. Novice evaluators tend to think of usability heuristics as a checklist rather than an inspiration. Thus, they become very focused on the heuristics and pay less attention to the system itself. This approach hinders their ability to fully experience the system and might lead them to detect false positive usability problems. On the other hand, once they find a usability issue that is related to a certain heuristic, they may rule out that heuristic and move on to the next even though they may still find other issues that are related to the same heuristic. Therefore, evaluators should perform two separate sessions. The first session should be a free evaluation in which the evaluators examine the system as if they were real users and do not consider the heuristics while performing the inspection. In the second session, the heuristics should be used to ensure that the evaluators did not miss any issues.
8. When examining different systems and websites, we can see that designers make specific mistakes. Providing a list of common mistakes to the evaluators and linking them to the heuristics could help the evaluators inspect the system. This could serve as a double-check procedure in which, after the evaluators finish the inspection, they could look at these mistakes and double-check whether they exist in the system that is being evaluated.
9. While all heuristics are important, some heuristics are associated with more usability issues than others. There is no agreement regarding which heuristics are associated with more usability issues, but there is agreement that there are discrepancies. Moreover, some usability heuristics are associated with more severe issues than others. Analyzing previous evaluations and providing the evaluators with insight into which heuristics require focus could be helpful, especially when the evaluation is performed under pressure to submit the results as quickly as possible. Ideally, the evaluators should focus equally on all the heuristics, but providing such information may be useful when the schedule is tight.
10. When performing the inspection, one session should be performed without depending on the heuristics, and the other session should be based on the heuristics. In the heuristic-based inspection session, the evaluator should go through each heuristic to determine its components, attempt to focus on the aspects that the heuristic addresses, and then inspect it based on whether there is a problem.
11. Assigning usability issues to a specific heuristic is an issue for novice evaluators. While some think that the most important aim is to identify usability problems regardless of whether they are assigned to a certain heuristic, others think that it is important, especially for novice evaluators, to assign such issues to a certain heuristic. The reason is that assigning usability issues to a specific heuristic increases the credibility of novice evaluators and the legitimacy of the report that

they submit to the developers. However, the task is sometimes confusing for two main reasons. First, novice evaluators tend to think that a single usability issue should be assigned to only one heuristic, while in fact, one issue can sometimes violate multiple heuristics. Second, it is difficult to determine how the usability problem is related to any of the heuristics. To facilitate this process, it is necessary to first examine the effect of the usability issue in light of the seven stages of action and usability components. When the stage of action affected by the usability issue is known, a number of heuristics are automatically eliminated. Moreover, when the usability component affected by the usability issue is known, other heuristics can be eliminated as well.

12. Accurately rating the severity of usability issues is a difficulty that novice evaluators face. Sometimes they overestimate the severity of the issue, while at other times, they underestimate it. To facilitate the rating of such issues, evaluators are encouraged to think of them in light of both the usability components and the seven stages of action. This approach will help in accurately rating the severity of an issue.
13. When reporting usability issues, in addition to what others have mentioned in previous studies [31], increasing the credibility and understandability of novice evaluators' reports is an issue. Two main solutions are suggested. First, explain the issue in light of the seven stages of action and how this issue hinders users in accomplishing their goals, and in light of how the issue affects a specific component of usability. Second, research is needed to support such claims with empirical results, similar types of problems, articles, psychological principles, etc.

4.2 Coherent Heuristic Evaluation (CoHE)

We can divide the CoHE process into three stages: understanding, inspecting, and documenting. In the first stage, the goal is to enable the evaluators to understand the main reasons for performing an evaluation and to understand the set of usability heuristics. In the second stage, the goal is to help the evaluators perform the evaluation. The aim of the third stage is to help the evaluators accurately and clearly describe the results of the evaluation.

Understanding

- a. Start by reading Norman's seven stages of action to understand how people go about accomplishing their goals.
- b. Read the definitions and explanations of the concept of usability, specifically Nielsen's definition of usability.
- c. Read the usability heuristics, and for each heuristic, read the description, example from everyday life, examples from the interface, different components, different aspects, significance, and when it is not applicable.
- d. Read how the heuristics are related to Norman's seven stages of action to understand the role each one plays in facilitating the accomplishment of the user goals.
- e. Read how the heuristics are related to the components of usability to understand the role each one plays in facilitating the usability of the system.

Inspecting

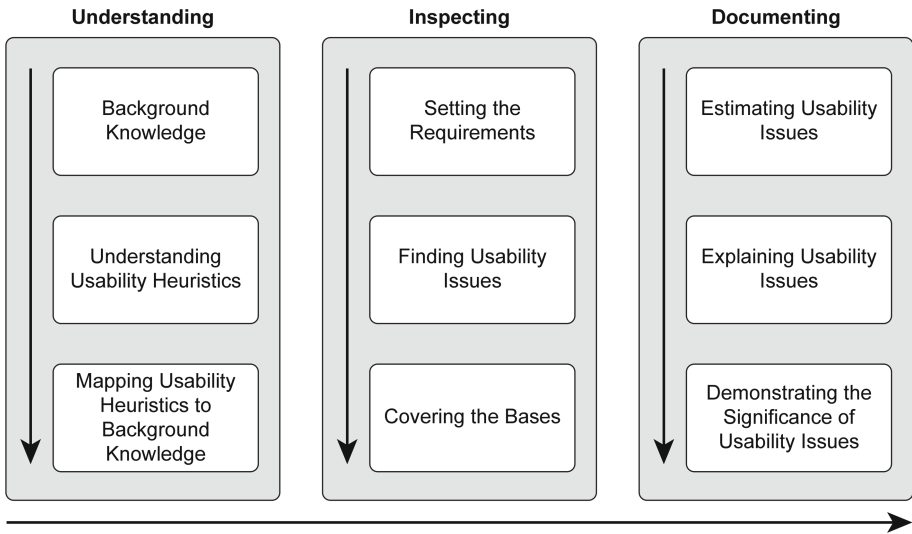
- a. Learn about what the system is, what it does and who the target audience is.
- b. Define the goals that users would want to accomplish by using the system.
- c. Before inspecting the system, allocate 30–45 min for each session. If a session would take more than 45 min, it is better to divide it into smaller sessions.
- d. First, inspect the system without referring to the usability heuristics. Attempt to accomplish the predetermined goals on the system. In addition, go to the homepage, try to learn more about the system from the about us page, and try to use the contact us page.
- e. Ideally, perform the inspection without the heuristics twice: once when not fully focused (e.g., tired or sleepy) to resemble users in a normal situation and once when fully focused and more critical. It is better to perform the former evaluation first to avoid the learning effect.
- f. Write down each usability issue without assigning it to any heuristic, and do not rate its severity.
- g. Second, inspect the system based on the heuristics. This time, examine the system, but when accomplishing the goals, examine the aspects that each heuristic addresses to evaluate whether they comply with the heuristic.
- h. Write down each usability issue without rating its severity.
- i. Finally, review the list of common problems associated with each heuristic and then see whether the system you are evaluating has similar problems. If so, write them down.

Documenting

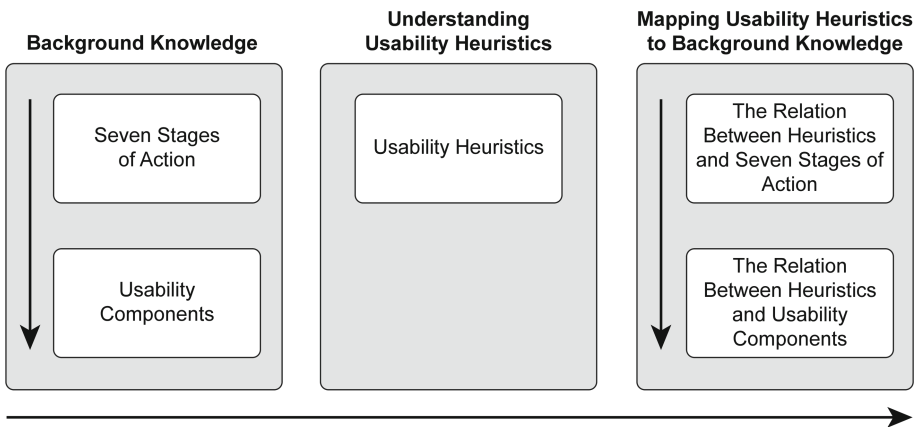
- a. For each usability issue found, start by assigning it to a usability heuristic or to multiple heuristics. To do so, identify which stage of action it affects and then eliminate the heuristics that are not associated with this stage. Then, determine which usability components it affects and eliminate all heuristics that do not deal with these components.
- b. For each usability issue, think of it in light of the usability components and the seven stages of action, and determine how it will affect the user in terms of accomplishing the goal and how it will affect the usability components.
- c. For each usability issue, write a description of the problem, take screenshots of it, and provide a recommendation on how to fix it.
- d. When describing the usability problem, try to explain it in light of the seven stages of action and the usability components.
- e. It is recommended that the description be supported by empirical studies, articles and descriptions of similar problems when possible.

The following diagrams (Diagram a, b, c, and d) provide a high-level overview of CoHE:

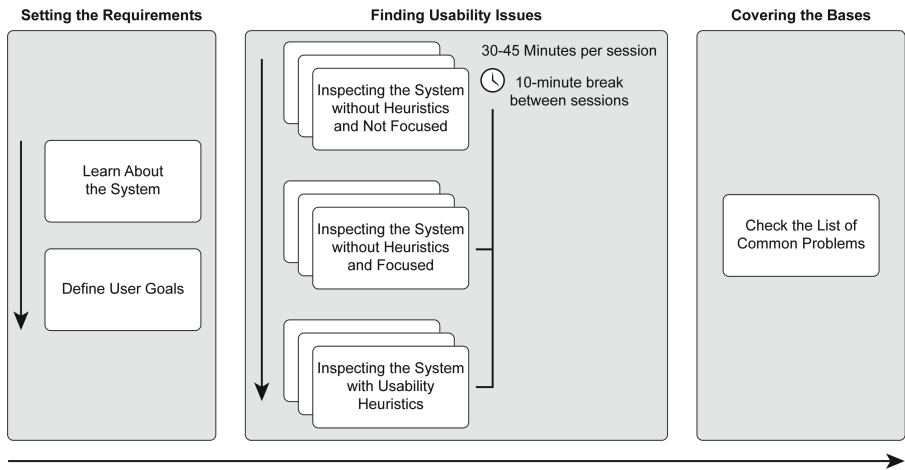
a. Stages of CoHE



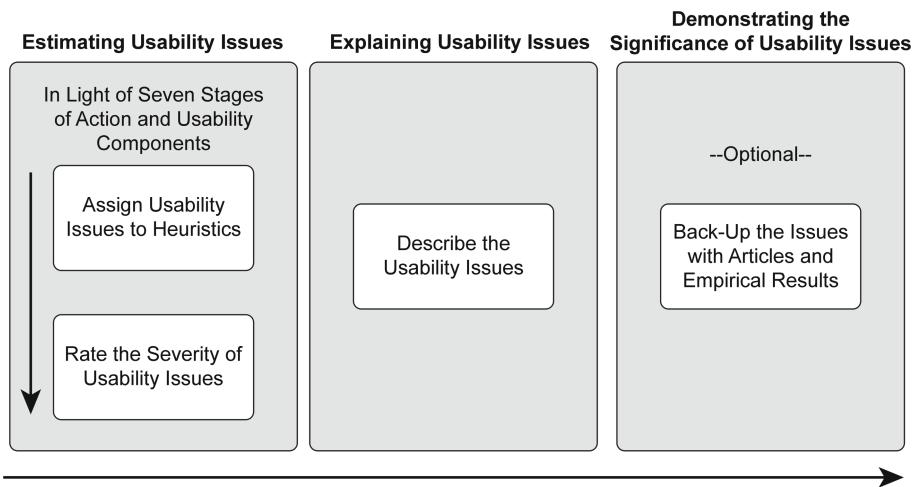
b. The Understanding Stage



c. The Inspecting Stage



d. The Documenting Stage



5 Discussion

HE is a discount usability method, meaning that it does not require many resources and is quick to apply. Admittedly, asking evaluators to read more about human ways of accomplishing goals and usability and to perform the evaluation sessions multiple times (based on their mood and on the heuristics) and asking them to put more effort

into supporting their results in the report will take more time. However, since novice evaluators tend to produce poor-quality results to the extent that their results are sometimes not very convincing to the developers, the procedure needs improvement. CoHE is a gateway to enhancing and mastering HE. It is expected that after performing CoHE multiple times, evaluators will gain more experience in both the knowledge side and the technique side of how to perform HE. For example, an evaluator will not need to read the seven stages of action and their relation to the heuristics after performing HE a few times because they will understand these concepts. Additionally, the duration of each session will decrease as the evaluators gain experience and become more confident and comfortable with performing HE. Therefore, the need to perform CoHE will eventually subside, but in the beginning, it is needed to increase the confidence of novice evaluators by reducing their confusion. Moreover, CoHE could be partially used as needed. For example, if an evaluator feels that he/she already knows how to handle certain parts of an evaluation, then he/she can skip that part. Although this version is meant for novice evaluators, experts can benefit from it as well. UX/HCI as a field is filled with many activities; thus, usability experts might stop performing HE for a while and then return to it. Therefore, this coherent version could help refresh their memory. CoHE is not based on specific heuristics but is a general step-by-step protocol that can be applied to any set of heuristics, either general heuristics or domain-specific heuristics. However, further work is needed in applying it to a specific set of heuristics and then validating its value.

6 Conclusion

HE is an inspection-based method in which a number of evaluators, typically 3–5, inspect a system based on a set of usability guidelines to identify usability issues. HE is one of the most common usability evaluation methods. Although it is very popular, it suffers from a number of problems. Studies show that the quality of HE when performed by novice evaluators is poor. This means that there is a discrepancy between the quality of expert results and novice results. To bridge that gap, we interviewed 15 usability experts to ask how they performed HE, the difficulties they faced and how they overcame them, and their suggestions on how to improve HE for novices. Based on their responses, we developed a list of problems, solutions and suggestions. From that list, we created a step-by-step protocol for HE to be used by novices that we call coherent heuristic evaluation (CoHE).

In future work, we plan to pursue three directions. First, we will apply CoHE to a specific set of heuristics to map usability problems to heuristics and to Norman's seven stages of action and usability components, explain each heuristic's aspects and components, check for common problems associated with each heuristic, and determine which heuristics detect the most issues. Second, we will validate CoHE by comparing it to traditional HE. Third, we will examine the applicability of CoHE to a wide range of sets of usability heuristics, both general heuristics and domain-specific heuristics.

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